Cascaded Convolution-Based Transformer With Densely Connected Mechanism for Spectral–Spatial Hyperspectral Image Classification

Baokai Zu, *Member, IEEE*, Yafang Li, Jianqiang Li, Ziping He, Hongyuan Wang, and Panpan Wu

***Abstract*— Hyperspectral image (HSI) classification attempts to classify each pixel, which is an important means of obtaining land-cover knowledge. HSIs are cubic data with spectral–spatial knowledge and can generally be considered as sequential data alongside spectral dimension. Unlike convolutional neural networks (CNNs), which mainly focus on local relationship models in images, transformers have been shown to be a powerful structure for qualifying sequence data. However, it lacks the excellent ability of CNNs in establishing local relationships in images and cannot perform good generalization in the case of insufficient data. In addition, the gradient disappearance problem hinders the convergence stability of deep learning networks as the layers get deeper. To address these problems, we propose a Cascaded Convolution-based Transformer with Densely Connected Mechanism (CDCformer) for HSI classification. First, we propose a cascaded convolution feature tokenization to extract spectral–spatial information, which will introduce some inductive bias properties of CNN into the transformer. In addition, we design a simple and effective densely connected transformer to enhance feature propagation and transfer memorable information from shallow to deep layers. It efficiently improves the performance of the transformer and extracts more discriminative spectral–spatial features from the HSI. Extensive experimental evaluation of three public hyperspectral datasets shows that CDCformer achieves competitive classification results.**

***Index Terms*— Cascaded convolution feature tokenization (CCFT), convolutional neural networks (CNNs), deep learning, densely connected mechanism (DCM), hyperspectral image (HSI) classification, transformer.**

I. Introduction

With the rapid development of remote sensing technology, it has become very easy to capture hyperspectral images (HSIs) for Earth observation. For decades, HSIs have received much attention because of the rich information contained in pixels. HSI has three dimensions, consisting of 1-D spectral features and 2-D spatial information, which includes spectral and spatial details of grounded objects. HSI pixels (spectral features) include several hundred narrow spectral bands, while spatial information includes the specific spatial distribution of land cover. With its rich information, HSI has been implemented in several fields, such as ecosystem monitoring [1], agriculture [2], and mineral exploration [3].

HSI classification attempts to identify the class of each pixel, which is an important means of obtaining land-cover information and has attracted much attention. In the past decades, a large number of hyperspectral (HS) classification techniques have been proposed [4], including k-nearest neighbor (KNN) [5], multinomial logistic regression [6], support vector machines (SVMs) [7], random forests (RFs) [8], and sparse representations (SRs) [9]. These methods can perform well on small sample classification problems. However, as the training size increases and the training data become more complex, these traditional methods often encounter performance bottlenecks due to limited data fitting and representation capabilities.

Recently, rapidly evolving deep learning techniques have accelerated the development of research areas such as image processing, speech recognition, and remote sensing [10], [11]. Deep learning employs multiple nonlinear layers to extract multiple robust features for classification, which can identify intrinsic and potentially valuable knowledge from big data. Owing to the power of nonlinear feature extraction, deep learning methods are widely employed for HSI classification [12]. Chen et al. [13] and Lin et al. [14] applied stacked autoencoders and deep belief networks (DBNs) [15] to extract the spectral–spatial features of HSI. These methods require flattening the image patches into a 1-D vector as the input. However, the spatial distribution of the original data is altered. With the popularity of convolutional neural networks (CNNs), CNN-based networks have been widely studied for HSI classification. Hu et al. [16] proposed a 1-D CNN network to capture the spectral features of HSIs. Compared with classical fully connected (FC) networks, CNNs are weight-sharing and locally connected, which makes their training process more efficient and effective. In [17] and [18], a 2-D CNN network is explored to exploit the spatial features of the local cubes. Unlike SAE, CNNs do not need to reshape the cube into

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a vector, thus maintaining as much spatial information as possible. Due to the small training sample size in high dimensions, it often causes overfitting problems. In [19] and [20], two branches of a 1-D and 2-D CNN structure are used to extract spectral and spatial features of pixels; these two features are then concatenated and fed to multiple FC layers to obtain joint spectral–spatial features. Hu et al. [21] proposed a two-stage hierarchical approach by using two cascaded CNNs. Zhong et al. [22] designed a spectral–spatial residual (SSR) model, which uses the original 3-D cube as an input to capture spectral–spatial features. Chen et al. [23] proposed a 3-D CNN approach to capture the effective features of HSIs. Luo et al. [24] have introduced a novel CNN framework, HSI–CNN, which combines 3-D CNN layers and 2-D CNN layers to study the features of HSIs. In [25] and [26], a hybrid spectral CNN (HybridSN) was proposed, which is first a spectral–spatial 3-D CNN followed by a spatial 2-D CNN. In addition, the complexity of using HybridSN is reduced compared to a pure 3-D CNN.

In the last few years, many promising approaches have combined CNNs and other theories. In [17], [23], and [27], principal components of the HSIs are extracted using principal component analysis (PCA) as the input to the CNN. Ghamisi et al. [28] proposed a self-improving model that adaptively selects the band with the maximal amount of information by fractional-order Darwinian particle swarm optimization. In [23], a virtual sample augmentation methodology was proposed to generate training data from the perspective of imaging procedures. A random zero setting was proposed by Kong et al. [29] to generate new labeled samples for the training of CNNs. In addition, transfer learning was used to solve the overfitting problem. A CNN trained from one HSI can be transferred to another image with only a few top layers to be fine-tuned to achieve satisfactory results [19], [30]. More importantly, the work in [31], [32], [33], and [34] showed that CNNs pretrained from nature images can be directly used to retrieve spatial features of HSIs. Wu and Prasad [35] proposed a semisupervised network for HSI classification that uses a limited set of labeled samples and a large number of unlabeled samples.

In deep networks, the residual connectivity mechanism has been shown to be an effective mechanism [22], [24], [36], [37]. In [36], a dense CNN approach was proposed to capture the features of HSI. An attention-based adaptive residual network (A^2S^2K-ResNet) has been proposed to capture the features of HSIs [37].

In addition to CNN, several other well-performing networks have been applied to HSI classification. Deng et al. [38] used a hierarchical and stacked sparse autoencoder to extract HSI features using only minimal labeled training data. In [39], a compact discriminative superposition autoencoder (CDSAE) was proposed which learns low-dimensional feature maps and progressively trains an effective classifier for HSI classification. Li et al. [40] described a DBN approach for HSI classification. A novel RNN model is proposed to effectively analyze HS pixels as sequence data and determine the categories by network inference [41]. Hang et al. [42] proposed an RNN framework to discover redundant knowledge of HSI through gated recursive units. In [43], a long short-term memory model for spectral–spatial HSI was proposed, which uses HSI pixels as data sequences to model the information correlation in the spectrum. In recent years, new network structures, such as generative adversarial networks (GANs) [44], [45] and graph convolutional networks (GCNs) [46], [47], [48], have been implemented to HSI classification to further improve the classification behavior. CNNs are always an indispensable and important module in the abovementioned network models.

HSIs are usually collected as cubic data with spectral–spatial knowledge, which can generally be considered as sequential data along the spectral dimension. The spectral information at different locations corresponds to different absorption properties at different wavelengths, which largely reflects the physical properties of the current object. Therefore, spectral features are very critical factors for accurate HSI classification. Although CNN-based methods have the advantage of extracting local spatial features, they are not good at modeling remote dependencies when dealing with sequential data. Recently, a new model called vision transformer (ViT) [49] has shown its superiority in processing sequential data and has achieved good performance in computer vision fields. The relationship between long-range spectral sequences of HSIs can be captured by the transformer. Although RNN-based models can also handle sequential data, they require sequential processing of the input sequences and have poor parallel computing capabilities. The feature extraction ability of transformers is superior to RNN models in which they can dynamically establish long-range dependencies of the input sequences. Several works have applied the transformer model to HSI classification. In [34], a spatial–spectral transformer (SST) network was proposed to construct relationships between neighboring spectra using a dense net of transformers. Qing et al. [50] presented SAT Net, which applies spectral attention and self-attention (SA) mechanisms to capture information from HSI images. Hong et al. [51] proposed a novel backbone called SpectralFormer, which learns spectral knowledge from grouped adjacent bands and proposes a module called cross-layer transformer (CAF). Recently, many new methods for HSI classification based on fusion ViT and CNN have been proposed. Zhang et al. [52] proposed convolution transformer mixer (CTMixer). In [53], a feature tokenization transformer (SSFTT) methodology was proposed to exploit the high-level semantic features of HSI. A group-aware hierarchical transformer (GAHT) confines MHSA to the local spatial–spectral context by introducing a new grouped pixel embedding (GPE) module [54]. The HSI transformer (HiT) [55] classification network captures the subtle spectral discrepancies and conveys local spatial context information by embedding convolution operations into the transformer structure.

An SA-based transformer [56] was used to capture global correlations from token sequences. However, most of the above methods refine the transformer approach by spectral feature processing. Although the transformer excels in capturing spectral features, it lacks image-specific inductive bias, such as localization and translation equivariance, compared to CNNs.

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Therefore, we propose a Cascaded Convolution-based Transformer with Densely Connected Mechanism (CDCformer) model for HSI classification. The model not only utilizes the inherent ability of the transformer to capture global information and models the correlation between adjacent sequences but also provides the CDCformer with the ability to explore local information by introducing the cascaded convolution feature tokenization (CCFT) module. In this model, first, the CCFT module is used to extract spectral–spatial features. As a result, some inductive bias inherent to CNNs can be introduced into the transformer, which gives the transformer the ability to explore local information as well. In addition, a densely connected transformer (DCT) is designed to enhance the information exchange between the layers and alleviate the loss of valuable information during layer-by-layer propagation. By learning to fuse information across layers and transferring memorized information from shallow to deep layers, we can significantly improve the performance of the transformer approach and extract discriminative information from the data. Finally, a softmax-based linear classifier is used to identify the labels of HS pixels.

The main contributions are summarized as follows.

1) A novel CCFT is proposed to extract spectral–spatial features of HSIs. It consists of three 3-D convolution layers, two 2-D convolution layers, and two FC layers. The obtained spectral–spatial features are then used as input tokens for the transformer encoder, which will introduce some of the inductive bias inherent to CNNs into the ViT while maintaining the advantages of transformers.
2) A simple and efficient DCT is designed to enhance feature propagation and transfer memorable information from shallow to deep layers. It can effectively improve the classification performance of the transformer and extract more discriminative information from the data.
3) A novel HSI classification model, CDCformer, is proposed by combining cascaded 3-D/2-D convolution feature tokenization and DCT, which significantly improves the classification accuracy. Evaluations on three typical datasets validate the effectiveness of the CDCformer model.

The rest of this article is structured as follows. Section II describes the CDCformer model. Section III conducts extensive experiments on the experimental datasets, designs the experimental parameters, and compares the classification accuracy. In Section IV, the conclusions are presented.

II. Proposed Method

This section first presents the overall framework of the CDCformer architecture. Next, we review some preliminary knowledge about classical transformers. Then, we further propose the CCFT model, which can significantly improve the performance of the transformer model. In addition, we propose a DCT. Finally, we briefly describe the effective design of its specific implementation.

A. Framework of the CDCformer Model

The proposed CDCformer model is shown in Fig. 1. From Fig. 1, we observe that the HSI classification framework is composed of a cascaded CNN, a DCT, and a multilayer perceptron (MLP). In this approach, a well-designed cascaded CNN is employed to capture the feature of HSI; then, a DCT is used to capture the association between the spectral sequences of HSI. Finally, an MLP is applied to obtain the classification results. These three components form a unified network and are trained with the support of training samples.

In CNNs, locality and translation equivariance are baked into the entire model. Although the SA layer is global, the ViT is very specialized in establishing long-range connections but with much less image-specific inductive bias compared to CNN. Therefore, we propose a CCFT architecture, which introduces the desirable features of CNN into the ViT while maintaining the advantages of a transformer.

At the same time, we design a DCT to establish layer-to-layer information exchange and mitigate the loss of valuable information during layer-by-layer propagation. The densely connected mechanism (DCM) can enhance feature propagation and transfer memory information from shallow to deep layers, thus effectively improving the performance of the transformer model.

B. Vision Transformer Module

Recently, ViTs [49] seem to reach or approach state-of-the-art results based on CNNs for various tasks in the vision domain since a typical transformer accepts a 1-D sequence embedding as input. The image $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ is partitioned into patches $\mathbf{x}_p \in \mathbb{R}^{N\times(P^2\cdot C)}$, where H, W, and C are the height, width, and channel of the image, (P, P) is the resolution of the patch, and $N = HW/P^2$ is the patches number obtained. These linear sequences of patch embeddings with D_l dimensions are provided to the transformer as input. Image sequence embeddings are processed in the same way as tokens (words) in NLP applications.

Each token sequence is given by $\mathbf{x}_p = [\mathbf{x}_p^1, \mathbf{x}_p^2, \ldots, \mathbf{x}_p^N]$. After adding the learnable embedding and the position embedding, the resulting $\mathbf{z}_0$ is sent to the transformer.

First, a learnable embedding is placed into the token sequences ($\mathbf{z}_0^0 = \mathbf{x}_{\text{class}}$), whose output status in the transformer encoder ($\mathbf{z}_L^0$) is represented as an image representation $\mathbf{y}$ [see (4)]. Here, L is the number of transformer encoder blocks. A classification head is connected to $\mathbf{z}_L^0$ during pretraining and fine-tuning. This classification head is implemented by a single linear layer of MLP.

The position embedding is then added to the token embedding to preserve the positional information. Here, we use a typical learnable 1-D position embedding. The positional information $\mathbf{E}_{\text{pos}}$ is then added to the token representation. The resulting sequence of embedding tokens is denoted by

$$\mathbf{z}_0 = \left[\mathbf{x}_{\text{class}}\,; \mathbf{x}_p^1\mathbf{E}; \mathbf{x}_p^2\mathbf{E}; \cdots; \mathbf{x}_p^N\mathbf{E}\right] + \mathbf{E}_{\text{pos}} \quad \mathbf{E} \in \mathbb{R}^{(P^2\cdot C)\times D_l}, \quad \mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1)\times D_l}. \tag{1}$$

Afterward, the obtained token sequences are fed into the transformer model. The transformer encoder [56] composed of multiple layers of multihead SA (MSA) blocks and MLP blocks [see (2) and (3)]. The layernorm (LN) is then used

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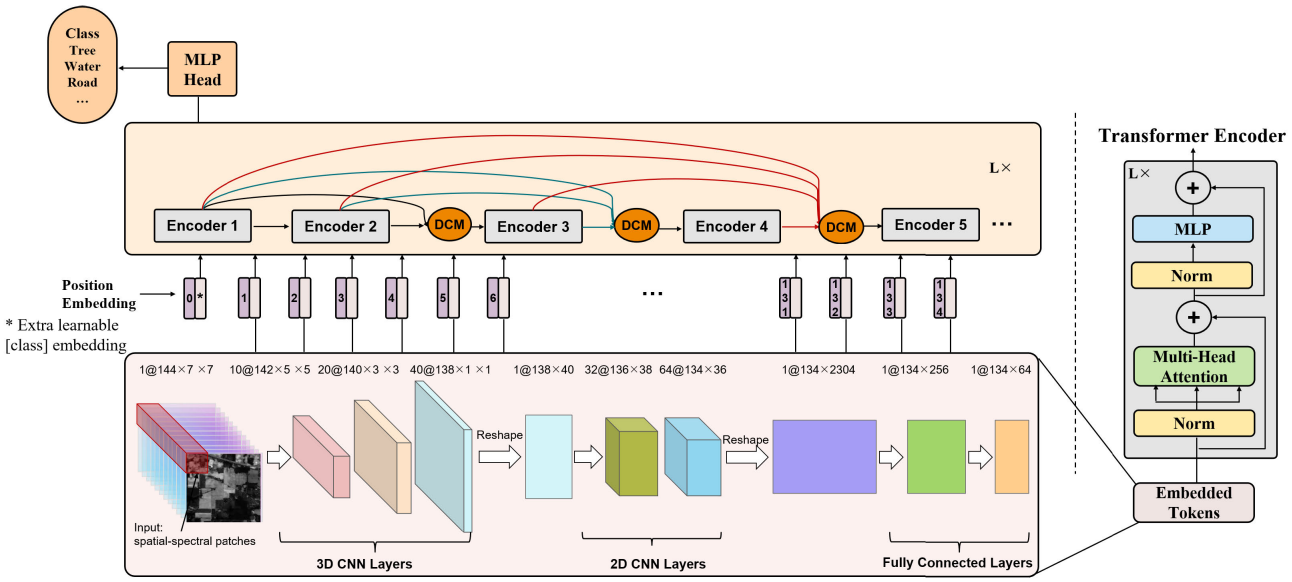


Fig. 1. Overall framework of the proposed CDCformer network for HSI classification.

ahead of MSA and MLP blocks, while the residual concatenation is utilized after each block [57], [58]. The MLP consists of two layers of Gaussian error linear unit (GELU). The forward calculation procedure is described as follows:

$$\mathbf{z}'_\ell = \mathrm{MSA}(\mathrm{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \ell = 1, \ldots, L \tag{2}$$

$$\mathbf{z}_\ell = \mathrm{MLP}\big(\mathrm{LN}\big(\mathbf{z}'_\ell\big)\big) + \mathbf{z}'_\ell, \quad \ell = 1, \ldots, L \tag{3}$$

$$\mathbf{y} = \mathrm{LN}\big(\mathbf{z}_L^0\big). \tag{4}$$

The transformer structure performs well due to its core MSA module. The usage of the SA effectively captures the correlation between sequences. To learn various implications, three learnable weight matrices, $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$, are determined. Through these three learnable weight matrices, the tokens are also linearly mapped to the 3-D invariant matrix of queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$. The attention scores are computed using the full contents of $\mathbf{Q}$ and $\mathbf{K}$, and the weights of the scores are computed with the softmax function. In summary, the expression of SA is

$$\mathrm{SA} = \mathrm{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_K}}\right)\mathbf{V} \tag{5}$$

where d_K is the dimension of $\mathbf{K}$.

The MSA block includes several weight matrices for mapping $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V}$, which are used to compute the MSA values using the same operational procedure. Then, the results of each attention head are concatenated. Finally, a linear transformation is performed to obtain the MSA results. The procedure is given as follows:

$$\mathrm{MSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{Concat}(\mathrm{SA}_1, S\ A_2, \ldots, \mathrm{SA}_H)\mathbf{W} \tag{6}$$

where H is the heads' number, $\mathbf{W} \in \mathbb{R}^{H \times d_K \times d_w}$ is the parameter matrix of the linear layers, and $d_w = N + 1$ is the tokens' number.

Next, the feature matrix obtained from the previously learned MSA module is fed to the MLP module. The MLP is followed by an LN, which enhances the gradient explosion, reduces the gradient vanishing problem, and makes the training faster.

For transformer encoder modules, the input $\mathbf{z}_{\ell-1}$ and the output $\mathbf{z}_\ell$ of the ℓth transformer encoder are equal in size. The learnable embedding $\mathbf{z}_L^0$ is fed to the top MLP for resulting classification. After the MLP, the likelihood of an input belonging to a particular class is computed by the softmax function.

C. Cascaded Convolution Feature Tokenization

Although the transformer is good at capturing spectral features, it lacks image-specific inductive biases, such as localization and translation variance, compared to CNNs. Therefore, the spectral–spatial features are first extracted using the convolution module; in this way, some inductive biases inherent to CNNs can be introduced into the transformer, which gives the transformer the ability to mine local information.

Since the HS pixels are cubic data or 3-D data, we extract the features of each pixel considering its spectral features and spatial information. Therefore, the cubic data are first fed into the convolution of the 3-D kernel, which can make more complete use of the spectral and spatial information hidden in the original data. Here, we use an appropriate window patch size of 7. Because if the window is too large, it increases the computational complexity, while, if the window is too small, the spatial information is not sufficient. Therefore, we adopt three 3-D convolutional layers, after which the spatial dimension becomes exactly 1. After reshaping, it is fed to the 2-D convolutional layers.

Since each feature vector is convolved from the same original data using different convolution kernels, which makes these vectors have different representations for specific features, there is a strong correlation between these vectors. All the original vectors are directly stitched into a matrix that has a constant height and whose width varies from the original

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TABLE I
CONFIGURATIONS OF THE PROPOSED CCFT. IT IS COMPOSED OF THREE 3-D CONVOLUTIONAL LAYERS (conv 3 d1, conv 3 d2, AND conv 3 d3), TWO 2-D CONVOLUTIONAL LAYERS (conv 2 d1 AND conv 2 d2), AND TWO FC LAYERS (FC1 AND FC2)

layers	Kernel size	output channels	operations
conv 3 d1	$3 \times 3 \times 3$	1/10	ReLU
conv 3 d2	$3 \times 3 \times 3$	10/20	ReLU
conv 3 d3	$3 \times 3 \times 3$	20/40	ReLU
conv 2 d1	3×3	1/32	ReLU
conv 2 d2	3×3	32/64	ReLU
fc1	–	2304/256	ReLU
fc2	–	256/64	ReLU

vectors to the number of feature vectors. After the reshaping operation, these data can be fed into the 2-D convolutional layers. In this way, the scarce HSI data correspond to exponential growth, which allows the model to obtain good performance and avoid overfitting.

Finally, to match the input of the transformer, we finally send it to the FC layers to obtain the embedding tokens for each pixel. Since the dimensionality is relatively large at the beginning, we use two FC layers to reduce it to a suitable dimensionality.

Therefore, we propose a well-designed cascaded convolution network that consists of three 3-D CNN layers with $3 \times 3 \times 3$ kernel size, two 2-D CNN layers with 3×3 kernel size, and two FC layers in turn. Table I illustrates the detailed configuration of the convolution layers.

Given the original HS data, $\mathbf{I} \in \mathbb{R}^{m \times n \times h}$, where $m \times n$ is the space size and h is the number of spectral bands. In $\mathbf{I}$, each pixel has h spectral bands and forms $Y = (y_1, y_2, \ldots, y_{Ca}) \in \mathbb{R}^{1 \times 1 \times Ca}$ as a one-hot category vector, where Ca is the land-cover category number. Therefore, the HSI is composed of h bands carrying rich spectral information.

First, we preprocess each original HSI pixel $\mathbf{I}$ into a 3-D patch $\mathbf{x}_P \in \mathbb{R}^{s \times s \times h}$, where $s \times s$ denotes a 7×7 window size. The position of the central pixel for each patch is defined as (x_i, x_j), with $0 \leq i < m$ and $0 \leq j < n$. Each patch's label is defined by the central pixel. The filling operation is performed on edge pixels because the adjacent pixels cannot be retrieved. The padding width on the image is $(s-1)/2$. As a result, we obtain $m \times n$ 3-D patches generated by $\mathbf{I}$. By removing the zero-labeled pixel patches, all the remaining sample patches were divided into a training set and a testing set.

After that, the spectral–spatial features of each pixel are evaluated using the CCFT. Each training sample is fed to the cascaded convolution layers. In this model, the three 3-D convolutional layers consist of k_1, k_2, and k_3 kernels with size $3 \times 3 \times 3$. Finally, k_3 spectral feature maps with size $h_1 \times 1 \times 1$ are obtained, where $h_1 = h - 6$. Therefore, after three 3-D convolution layers, we obtain k_3 feature maps, i.e., $k_3@h_1 \times 1 \times 1$.

Each feature map is obtained from a different convolution kernel, which makes these specific feature maps have different representations for HSI. On the other hand, there is a strong correlation between these vectors. Therefore, all feature maps are directly stitched into a matrix. After the rearrangement process, the feature map with size $1@h_1 \times k_3$ is sent to the next 2-D CNN layers.

For 2-D CNN layers, conv 2 d1 and conv 2 d2 have $k_4 = 32$ and $k_5 = 64$ kernels of size 3×3, respectively. The size of the output feature map after two 2-D convolutional layers becomes $k_5@h_2 \times s_1$, where $h_2 = h - 10$ and $s_1 = k_3 - 4$.

After that, the k_5 feature maps are rearranged into $1@l_2 \times s_2$, which is employed as the input to FC layers fc1 and fc2 with s_2 and D_l nodes, respectively. Here, $s_2 = k_5 \times s_1$. The spectral–spatial features $\mathbf{x} \in \mathbb{R}^{N \times D_l}$ of each pixel obtained from the CCFT module are used as embedding tokens for the transformer encoder after adding the learnable embedding and position embedding. Here, $N = h_2$ and $D_l = 64$.

D. Densely Connected Transformer

For deep learning models, there is a common problem of vanishing gradient, which hinders the convergence stability of transformer models. The skip connection (SC) mechanism has been shown to be an effective mechanism to enhance the exchange of information across layers and minimize information loss during network learning.

Therefore, inspired by the cross-layer adaptive fusion module [51], we design a DCT to fuse cross-layer information, instead of simply stacking transformer encoder blocks in series as in the standard ViT. Through the DCM, the shallow memory information is transferred to the deeper layers, which can effectively enhance the characteristics of the transformer model.

The proposed DCT consists of L transformer encoders. Consider that $\mathbf{e}_1 \in \mathbb{R}^{(N+1) \times D_l}, \ldots, \mathbf{e}_{\ell-1} \in \mathbb{R}^{(N+1) \times D_l}$ denotes the output of the previous transformer encoder blocks from block 1th to block ℓth. Then, the ℓth transformer block receives the outputs $\mathbf{e}_1, \ldots, \mathbf{e}_{\ell-1}$ from the previous layers.

To concretely implement the dense connection mechanism, we first use the unsqueeze method to increase the dimensions of the previous layers' output $\mathbf{e}'_1, \ldots, \mathbf{e}'_{\ell-1} \in \mathbb{R}^{(N+1) \times D_l \times 1}$. The unsqueeze method is used to change the dimensions of the tensor, by adding an element of size one at a specific location, which can be used to create another tensor as yield. Then, the outputs $\mathbf{e}'_1, \ldots, \mathbf{e}'_{\ell-1}$ of the previous transformer blocks are concatenated. After the concatenate operation, we obtain $\hat{\mathbf{e}}_{(\ell-1)} \in \mathbb{R}^{(N+1) \times D_l \times (\ell-1)}$. Then, a convolution of kernel $1 \times (\ell - 1)$ is applied to fuse the previous layers to obtain $\mathbf{e}'_{(\ell-1)} \in \mathbb{R}^{(N+1) \times D_l \times 1}$. Finally, the last dimension is removed by squeeze. This process can be defined as follows:

$$\mathbf{e}_\ell = \mathrm{Con}_\ell\left(\mathrm{concat}\left[\mathbf{e}'_1, \ldots, \mathbf{e}'_{\ell-1}\right]\right) \tag{7}$$

where $\ell \in 2, \ldots, L$. Con_ℓ is a convolution operator with kernel size of $1 \times (\ell - 1)$.

The DCT encourages feature fusion and minimizes the loss of valuable information during propagation. Fig. 2 shows the proposed DCT with $L = 5$.

E. Implementation

The CDCformer introduces the inductive bias inherent to CNNs into the transformer through CCFT. In addition, a DCT

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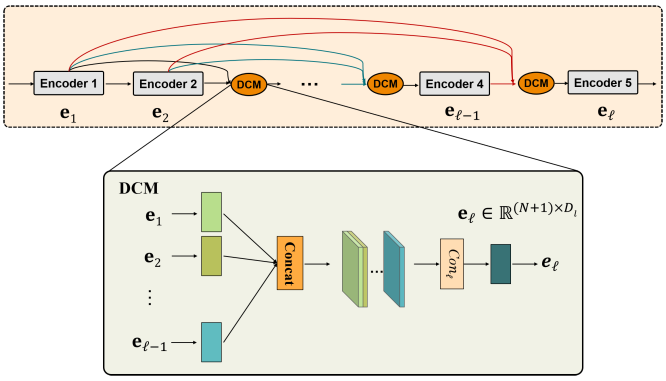


Fig. 2. Illustration of the proposed DCT.

is designed to encourage feature fusion and improve the stability of model convergence.

Here, we choose the Houston2013 Data with size 394 × 1905 × 144 as an example to demonstrate the proposed CDCformer model.

First, a 3-D patch is performed on the HSI dataset $\mathbf{I}$. The adjacent patch of each 3-D patch $\mathbf{x}_P \in \mathbb{R}^{7\times7\times144}$ is employed as the input data for the CNN layers. After 3-D convolution layers with 3 × 3 × 3 cubic kernel on each patch, 40 spectral feature maps 40@138 × 1 × 1 containing spectral–spatial information are finally obtained. After the rearrangement operation, the feature map's size becomes 1@138 × 40, which can be used as the input of 2-D CNN layers. After two 2-D CNN layers, the output feature size is 64@134 × 36. Then, the 64 feature maps are rearranged into 1@134 × 2304 and fed into the FC layers fc1 and fc2 with 256 and 64 nodes, respectively. Thus, after the CCFT module, the spectral–spatial features of each pixel are used as embedding tokens $\mathbf{x}_p \in \mathbb{R}^{134\times64}$ to ViT, where the number of tokens is $N = 134$ and the dimension of the tokens is $D_l = 64$.

A learnable embedding is first placed into the embedded token sequence before being fed into the transformer encoder. Then, the position embedding is added to the token embeddings. The obtained sequence of token embeddings is $\mathbf{z}_0 \in \mathbb{R}^{(N+1)\times D_l}$.

Here, the DCT consists of five transformer encoder blocks. The ℓth block receives the outputs $\mathbf{e}_1, \ldots, \mathbf{e}_{\ell-1} \in \mathbb{R}^{135\times64}$ generated by the previous blocks, where $\ell \in 2, \ldots, L$. For the DCM, the size of the outputs $\mathbf{e}'_1, \ldots, \mathbf{e}'_{\ell-1} \in \mathbb{R}^{135\times64\times1}$ from the previous layers is first increased by the unsqueeze method. Then, the outputs of the previous transformer blocks are concatenated to obtain $\hat{\mathbf{e}}_\ell \in \mathbb{R}^{135\times64\times(\ell-1)}$. Afterward, a convolution Con_ℓ of $1 \times (\ell - 1)$ kernel size is applied to fuse the information of the previous blocks and obtain $\mathbf{e}'_\ell \in \mathbb{R}^{135\times64\times1}$. Finally, we use the squeeze method and obtain the output $\mathbf{e}_\ell \in \mathbb{R}^{135\times64}$ of the ℓth block.

The learnable embedding $\mathbf{z}^0_L \in \mathbb{R}^{1\times64}$ is fed to the LN and linear classifier to obtain the labels for the judgments. The general procedure of the Houston2013 classification based on the proposed CDCformer model is shown in Algorithm 1.

Algorithm 1 Houston2013 Dataset Classification Based on the Proposed CDCformer Model

Input: HSI dataset $\mathbf{I} \in \mathbb{R}^{m\times n\times h}$ and the ground-truth $\mathbf{Y} \in \mathbb{R}^{m\times n}$, the train sample index $\mathbf{TR} \in \mathbb{R}^{m\times n}$ and the test sample index $\mathbf{TE} \in \mathbb{R}^{m\times n}$, patch size $s = 7$.
Output: The predicted label of test dataset.
1: Setup batch size as 64, Adam optimizer with $5e-4$ learning rate, 600 epochs ϵ.
2: Create training samples and testing samples from sample patches in the $\mathbf{I}$.
3: Build training loader and testing loader.
4: **for** $i = 1$ to ϵ **do**
5: Execute the 3-D convolution layers. And reshape feature map from 40@138 × 1 × 1 to 1@138 × 40.
6: Perform 2-D convolution layers. And Reshape feature map of 64@134 × 36 into 1@134 × 2304.
7: Perform the fully connected layers to generate embedding tokens.
8: Put the learnable embedding in the embedded token sequences and add the position embedding to the token embeddings.
9: Perform densely connected transformer module.
10: Fed the learnable classification embedding to the MLP layer.
11: Apply the linear layer to recognize the labels.
12: **end for**
13: Obtain the predicted labels using the testing dataset and the trained model.

III. Experiments

This section first describes three well-known HS datasets, followed by the experimental setup and comparison methods. Then, we perform an ablation analysis and a parametric analysis of the proposed model. Finally, extensive exercises are performed to analyze the capabilities of the CDCformer from quantitative and qualitative perspectives.

A. Data Description

During this study, the evaluation of the presented method was performed on three publicly available HSI datasets, including the Houston2013 Data, the Kennedy Space Center (KSC) dataset, and the Botswana dataset.

1) *Houston2013 Data:* The Houston2013 dataset was obtained by the ITRES CASI-1500 instrument at the University of Houston campus and its surrounding rural region in Texas, USA. The HS cube consists of 349 × 1905 pixels and 144 bands with wavelengths of 364–1046 nm at 10-nm intervals. It has been widely applied as a benchmark dataset to evaluate land-cover classification performance. The Houston2013 dataset that we use here is the cloud-free version.[1] Table II lists the 15 challenging land-cover types and the corresponding sample sizes for the training and testing sets. For the visualization the training, testing, and ground-truth

[1]The data were provided by Prof. N. Yokoya from The University of Tokyo and RIKEN AIP.

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TABLE II
LAND-COVER CLASSES OF THE HOUSTON2013 DATASET, AND THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot1	192	1041
13	Parking Lot2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
	Total	2832	12197

TABLE III
LAND-COVER CLASSES OF THE KSC DATASET, AND THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Scrub	38	723
2	Willow swamp	12	231
3	CP hammock	13	243
4	CP/Oak	13	239
5	Slash pine	8	153
6	Oak/Broadleaf	11	218
7	Hardwood swamp	5	100
8	Graminoid marsh	22	409
9	Spartina marsh	26	494
10	Cattail marsh	20	384
11	Salt marsh	21	398
12	Mud flats	25	478
13	Water	46	881
	Total	260	4951

TABLE IV
LAND-COVER CLASSES OF THE BOTSWANA DATASET, AND THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Water	27	243
2	Hippo grass	10	91
3	Floodplain grasses1	25	226
4	Floodplain grasses2	21	194
5	Reeds1	27	242
6	Riparian	27	242
7	Firescar2	26	233
8	island interior	20	183
9	Accacia woodlands	31	283
10	Acacia shrublands	25	223
11	Accacia grasslands	30	275
12	Short mopane	18	163
13	Mixed mopane	27	241
14	Exposed soils	10	85
	Total	324	2924

samples offered by the 2013 IEEE GRSS data fusion competition[2].

2) The KSC dataset was captured in 1996 from the AVIRIS instrument over KSC, Florida, in the wavelength range of 400–2500 nm. These images are spatially sized at 512 × 614 with 176 spectral bands, where water absorption and low signal-to-noise ratio (SNR) bands are removed. It includes 5202 samples from 13 upland and wetland categories. Here, 5% of the overall samples were randomly selected from the KSC dataset as the training set. The names of the land-cover categories, the training samples, and the test samples are listed in Table III.
3) The Botswana dataset was collected by the NASA EO-1 satellite over the Okavango Delta in Botswana in 2001. The sensor on EO-1 obtained data in 242 bands at a pixel resolution of 30 m over a 7.7-km strip, which covered the 400–2500 nm fraction at 10-nm intervals. Uncalibrated and noisy bands covering water absorption were removed, leaving 145 bands as feature candidates. The available data analyzed here include 14 defined categories representing observations of land-cover categories. Here, 5% of the samples from the Botswana dataset were randomly selected as the training set. These samples are listed in Table IV.

B. Experimental Setup

1) Evaluation Metrics: To quantitatively analyze the classification performance, four quantitative evaluation metrics were introduced, namely, overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and classification accuracy per category. In addition, the classification maps acquired by various models are visualized for qualitative comparison.

2) Configuration: The validation experiments of the CDCformer method were implemented in a PyTorch environment using a 12th-generation Intel[3] Core[4] i9-12900K CPU with 3.19 GHz, 64-GB RAM, and an NVIDIA GeForce RTX 3090 24-GB GPU. For batch training, each minibatch size was chosen to be 64. Using Adam's Optimizer as the optimizer, the initial learning rate was defined as $5e-4$, which was reduced by 0.97, 0.94, and 0.94 for the Houston2013, KSC, and Botswana datasets, respectively. Moreover, the training epoch is set to 600 roughly. However, in some datasets, the CDCformer was able to converge using fewer epochs.

For CDCformer, the patch size is set to 7 × 7. The embedded tokens have 64 dimensions and are fed to the transformer encoder blocks. The number of heads and the hidden size of the MLP are the same as in SpectralFormer [51]. Each encoder block includes a four-head SA layer, an MLP with eight hidden neurons, and a GELU [59]. A dropout layer was employed after the MLPs for dropping 10% neurons.

3) Comparison Methods: The proposed CDCformer is validated by using several comparison methods, including KNN, RF, SVM, 1-D CNN [60], 2-D CNN [23], SSRN [22], HybridSN [25], [26], RNN [42], SAE [13], ViT [56], and SpectralFormer [51], which are shown as follows.

[2]http://www.grss-ieee.org/community/technical-committees/data-fusion/fusion/2013-ieee-grss-data-futon-contest/

[3]Registered trademark.
[4]Trademarked.

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1) For KNN, the nearest neighbor number k is set as 5.
2) In the case of RF, 1000 decision trees are applied here.
3) The libsvm toolbox is chosen to implement the HSI classification task. The radial basis kernel function (RBF) is selected. In RBF, cross-validation is performed in the scope of $\sigma = [2^{-3}, 2^{-2}, \ldots, 2^4]$ and $\lambda = [10^{-2}, 10^{-1}, \ldots, 10^4]$ by fivefold cross-validation. After that, the best combination of hyperparameters σ and λ is obtained.
4) For the 1-D CNN, the basic network unit is a convolutional block consisting of a 1-D convolutional filter with 128 output size, a batch normalization (BN) layer, and an ReLU nonlinear function. Finally, a softmax function is attached to the top layer.
5) The 2-D CNN framework consists of three modules and a softmax function. Each module consists of a 2-D CNN layer, a BN, a max-pooling, and an ReLU. Each 2-D convolutional layer contains 32, 64, and 128 3×3 filters, respectively. The input to the 2-D CNN is a small cube of size $16 \times 16 \times h$, where h denotes the number of spectral bands in each dataset.
6) Considering that HSI contains one spectral dimension and two spatial dimensions, SSRN [22] proposed a framework for successive extraction of spectral and spatial features. By adding SCs between each layer, a hierarchical feature representation layer is formed for successive residual blocks. Specifically, the network consists of two spectral and two spatial residual blocks, an average pooling layer, and an FC layer. The input to the SSRN network is a small cube of size $7 \times 7 \times h$, where h denotes the number of spectral bands for each dataset.
7) The HybridSN [25], [26] is a spectral–spatial 3-D CNN followed by a spatial 2-D CNN. The 3-D CNN facilitates a joint spectral–spatial feature representation from the spectral band stacking. The 2-D CNN on top of the 3-D CNN further learns a more abstract spatial representation. To first eliminate spectral redundancy, PCA is performed on the raw HSI data along the spectral bands to obtain 30 components. The HybridSN framework contains three 3-D CNN layers, one 2-D CNN layer, flatten, and three FC layers. After the first two FC layers, there are two dropout layers with a dropout rate of 0.4. Here, the ReLU nonlinear function is adopted. The input to the HybridSN model is a cube of size $25 \times 25 \times 30$.
8) For the RNN, there are two recursive layers with a gated recurrent unit (GRU). The neurons in each layer are given as 128. The input to the RNN is the raw spectral information of each pixel.
9) The input of SAE is the raw spectral information of each pixel. There are three layers in total. The number of neurons is set to 32, 64, and 128. ReLU is used as the activation function.
10) For the transformer, in the ViT-based HSI model, there are five encoder blocks [51].
11) For the SpectralFormer [51], the structure is the same as the transformer described above. The size of the patches is empirically set to 7×7. More specifically, embedding tokens are fed to the transformer encoder for HSI classification. In each encoder block, a four-head SA layer is included, as well as an MLP with eight hidden dimensions, and a GELU [59] activation function. After encoding the positional embedding, a dropout layer was employed in the MLP to drop 10% neurons. An ℓ_2 weight decay regularization [61] with parameter $5e-3$ was also employed to prevent the risk of potential overfitting [51].

TABLE V
ABLATION ANALYSIS OF THE PROPOSED CDCFORMER ON THESE THREE DATASETS

Data sets	Cases	Module		Metric		
		CCFT	DCT	OA (%)	AA (%)	$\kappa \times 100$
Houston2013	Patch-wise ViT	✗	✗	83.81	84.36	82.42
	CCF-based ViT	✓	✗	86.67	88.05	85.53
	DCT-based ViT	✗	✓	84.55	83.27	83.23
	CDCformer	✓	✓	**90.57**	**91.51**	**89.76**
KSC	Patch-wise ViT	✗	✗	86.41	78.45	84.86
	CCF-based ViT	✓	✗	90.81	86.45	89.77
	DCT-based ViT	✗	✓	88.65	81.36	87.36
	CDCformer	✓	✓	**92.63**	**86.95**	**91.79**
Botswana	Patch-wise ViT	✗	✗	96.92	96.90	96.67
	CCF-based ViT	✓	✗	97.06	97.37	96.81
	DCT-based ViT	✗	✓	96.99	97.37	96.74
	CDCformer	✓	✓	**97.57**	**97.79**	**97.37**

C. Model Analysis

1) Ablation Analysis: We investigated the performance of CDCformer to improve the classification accuracy by progressively adding different modules (i.e., CCFT and DCT) to the network. For this purpose, we performed extensive ablation exercises on these three datasets to verify the efficiency of these modules (or components) in CDCformer for HSI classification, as listed in Table V.

In more detail, the classical transformer without CCFT and DCT modules produces the weakest classification results in terms of accuracy. By inserting CCFT or DCT in ViT, the classification results are superior to those of ViT. Even better, the joint utilization of CCFT and DCT can provide a huge improvement (more than 6% OA). This is because the spectral–spatial characteristics of HSI may be learned more efficiently and easily due to the joint use of CCFT and DCT.

2) Parameter Analysis: To verify the effectiveness and robustness of the spectral–spatial transformer, this section analyzes some key parameters involved in the transformer models, which include the depth of the transformer encoder, the number of MSA heads and learning rate, the patch size, the number of adjacent bands, and the size of training samples. Both the depth of the transformer encoder and the number of MSA heads affect the robustness and complexity of the model. The learning rate may affect the performance of the model. Therefore, the optimal parameter settings for these parameters need to be investigated. In addition, we perform a parametric analysis of the patch size, the number of adjacent bands, and the scale of training samples and compare them with other methods.

To analyze the effect of these parameters on the ViT-based models, the other parameters were kept fixed. For the depth of

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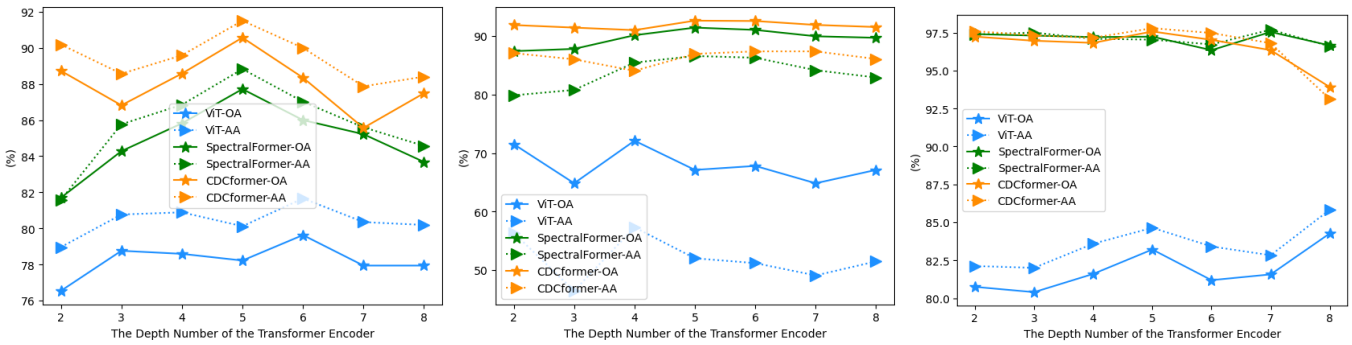


Fig. 3. Classification results with the varying depth number of the transformer encoder of ViT-based models on these three datasets. (a) Houston2013. (b) KSC. (c) Botswana.

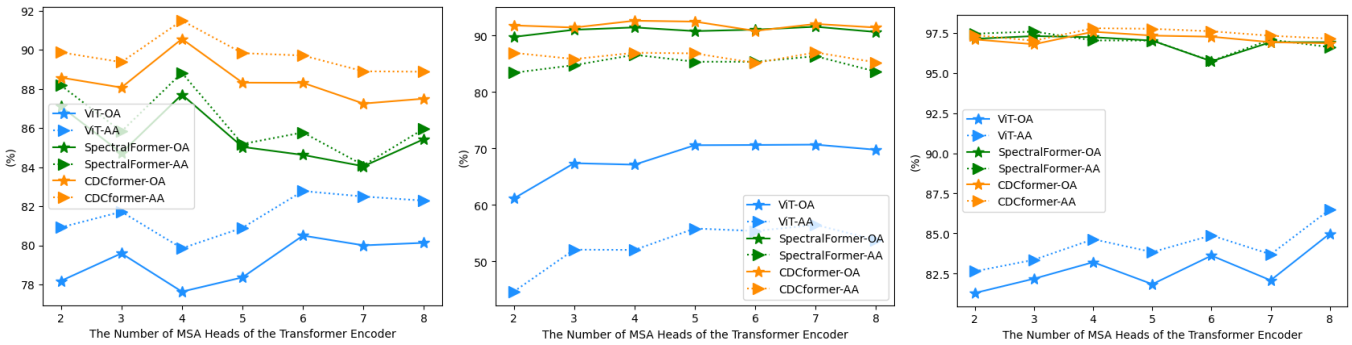


Fig. 4. Classification results with the varying number of MSA heads of the transformer encoder of ViT-based models on these three datasets. (a) Houston2013. (b) KSC. (c) Botswana.

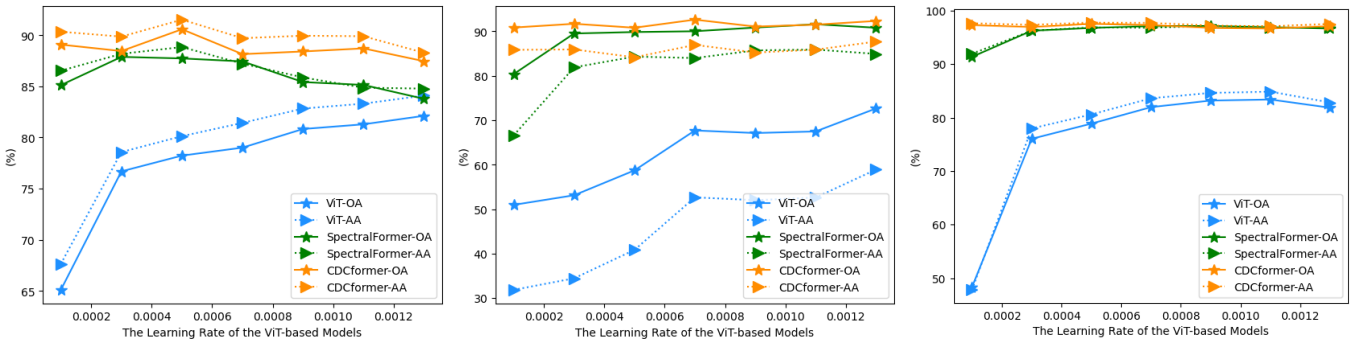


Fig. 5. Classification results with the varying learning rate of ViT-based models on these three datasets. (a) Houston2013. (b) KSC. (c) Botswana.

the transformer encoder, a search was performed from a depth of 2–8. Fig. 3 reflects the classification accuracy result for the three datasets; we also show the radar chart of the OA, which can help to demonstrate the effectiveness of the proposed method in the Appendix (see Fig. 11). From Figs. 3 and 11, we can observe that, for all depths of the CDCformer, pretty good classification accuracy was obtained. Similarly, for the other two models, depths play a small role in the classification results. However, the other two models are significantly worse than our model. In the experiments, based on these results, the depth of the transformer encoder is set to 5 to obtain better classification results.

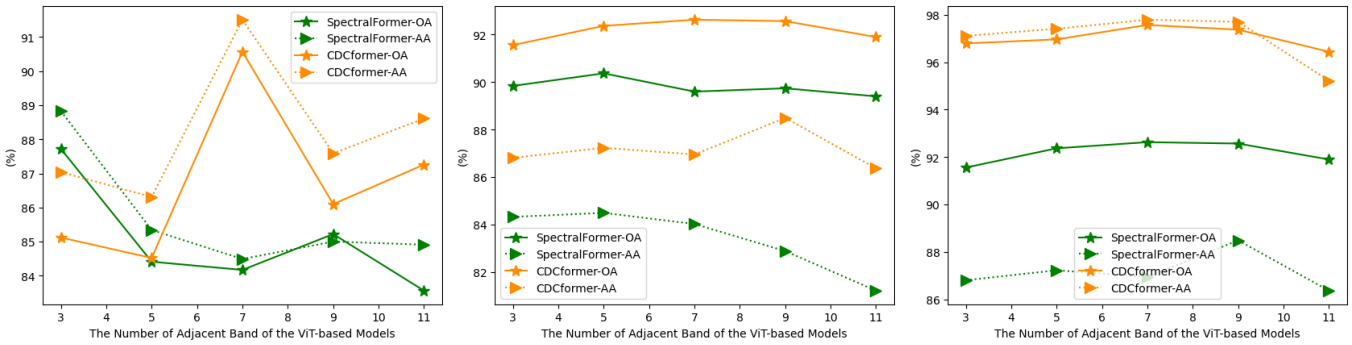
For the number of MSA heads, a search was performed from a depth of 2–8. Fig. 4 shows the classification accuracy results for the three datasets; we also show the radar chart of the OA in the Appendix (see Fig. 12). From Figs. 4 and 12, we can get similar conclusions to the depth of the transformer encoder experiment. We can observe that, for the number of other MSA heads, fairly good classification accuracy is obtained. The best results are obtained with a number of heads of 5, probably because the other hyperparameters are fixed.

To find the optimum value of the learning rate, we choose $1e-4$ to $1.3e-3$ with an interval of $2e-4$ for the search. Figs. 5 and 13 show the classification results for different learning rates. Within a certain range, the parameter is insensitive, which provides a great possibility for the proposed CDCformer model to be used in practical applications. In other

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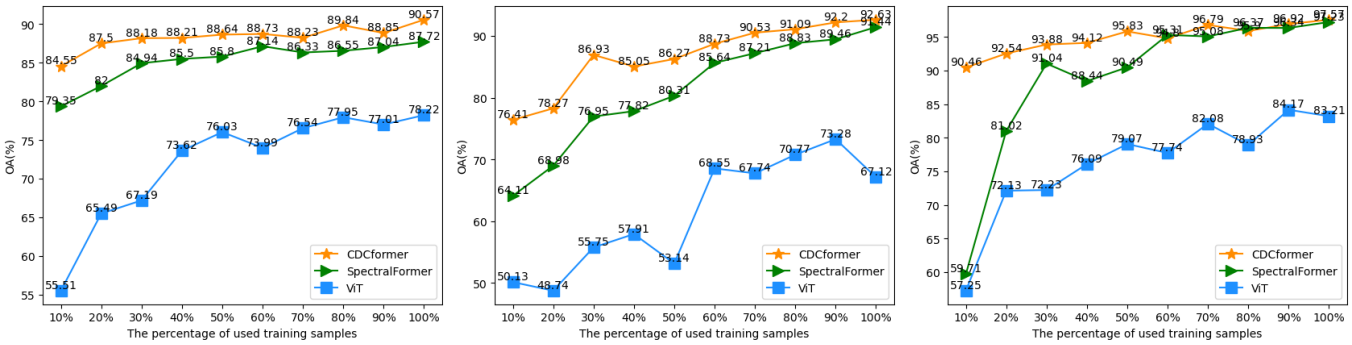
TABLE VI
PATCH SIZE ANALYSIS OF VIT-BASED MODELS ON THESE THREE DATASETS

Data sets		Houston2013			KSC			Botswana		
Models	Patch Size	7	9	11	7	9	11	7	9	11
ViT	OA (%)	78.22	82.37	80.40	67.12	87.92	86.14	83.21	96.07	95.04
	AA (%)	80.12	81.53	78.67	52.01	81.77	79.54	84.64	96.38	95.32
SpectralFormer	OA (%)	87.72	82.45	82.61	91.44	88.67	87.92	97.23	97.71	96.17
	AA (%)	88.83	82.56	81.41	86.59	80.34	79.77	97.03	97.88	96.11
CDCformer	OA (%)	**90.57**	86.15	85.39	92.63	91.80	**93.19**	97.57	97.37	**97.78**
	AA (%)	**91.51**	87.71	86.84	86.95	85.17	**87.80**	**97.79**	97.67	97.24



(a) (b) (c)

Fig. 6. Classification results with the varying number of adjacent bands of the SpectralFormer and CDCformer on these three datasets. (a) Houston2013. (b) KSC. (c) Botswana.



(a) (b) (c)

Fig. 7. Classification results (OA) obtained by ViT-based models with the varying number of used training samples on the Houston2013 dataset. (a) Training. (b) KSC. (c) Botswana.

words, this parameter can be simply and directly applied to other datasets.

We also analyze the patch's window size and the number of adjacent bands. Because of our cascade convolution feature tokenization network configuration, the spatial window size should be larger than 7. We choose the spatial window size as 7, 9, and 11. To adapt the 3-D convolutional output to the subsequent 2-D convolutions, we apply a 3-D adaptive average pooling on the 3-D convolutional output. Table VI shows the classification results for different spatial patch sizes. We can see that, for the KSC and Botswana datasets, the CDCformer is relatively stable as the patch size increases; however, for the Houston2013 dataset, the classification results decrease. This is due to the fact that the landscape distribution is fragmented. In the Houston2013 dataset, the spatial distribution of grasslands, roads, commercial, and so on is chaotic. Therefore, larger window patch sizes are not as effective as smaller ones. However, for the KSC and Botswana datasets, the landscape distribution is relatively complete and smooth. Therefore, the effect of patch size is not very obvious. We also analyzed the size of the spectral windows, i.e., the number of adjacent bands for $3, 5, 7, 9$, and 11. As seen in Figs. 6 and 14, the number of adjacent bands has an impact on the performance of the CDCformer. Combining the three datasets, the number of adjacent bands is chosen as 7, which is more suitable for CCFT.

In addition, we randomly selected the training datasets provided by the three datasets at $[10\%, 20\%, \ldots, 100\%]$ intervals. Here, the testing set is kept fixed. Fig. 7 reports the average results OA obtained on the ViT-based models. We can see that the classification performance gradually improves as the percentage of training samples used increases. Moreover, the proposed CDCformer significantly outperforms the other two ViT-based models. It is worth noting that OA achieves

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TABLE VII

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS ON THE HOUSTON2013 DATASET. "*," "**," AND "***" INDICATE THAT THE IMPROVEMENT OF THE CDCFORMER OVER THE BASELINE IS SIGNIFICANT AT THE 0.1, 0.05, AND 0.01 LEVELS, RESPECTIVELY

Class No.	KNN	RF	SVM	1-D CNN	2-D CNN	SSRN	HybridSN	RNN	SAE	ViT	SpectralFormer	CDCformer
1	79.01	**89.56**	81.48	85.36	84.90	83.51	72.52	82.79	83.31	81.86	84.71	81.58
2	77.35	73.14	97.74	95.77	96.39	99.12	89.94	95.35	97.85	95.77	97.74	**98.87**
3	97.23	98.02	99.60	99.92	89.15	98.61	79.56	99.41	**99.64**	99.60	96.63	93.86
4	81.16	82.13	87.22	93.38	96.12	98.71	76.87	96.48	96.86	**99.62**	98.67	97.25
5	91.00	89.40	97.25	96.31	97.28	99.98	99.65	95.11	96.91	96.78	**99.81**	98.48
6	90.91	93.71	95.80	96.43	92.73	95.10	93.36	96.36	**96.85**	95.10	94.41	95.80
7	76.87	79.01	79.01	83.46	90.86	92.5	63.26	83.57	81.59	80.04	83.12	**91.14**
8	55.37	64.57	75.50	50.78	59.53	85.93	73.72	54.16	50.34	58.02	82.91	**83.67**
9	71.48	75.54	77.34	63.47	78.16	81.98	70.72	74.07	71.80	69.41	74.69	**84.04**
10	56.66	39.01	58.40	72.52	**92.69**	70.81	55.24	74.96	81.54	49.71	82.72	91.99
11	66.51	60.67	78.18	83.43	83.48	95.43	75.76	81.28	75.95	80.17	84.35	**89.56**
12	76.37	74.03	84.25	68.46	**90.24**	84.94	89.03	68.24	68.27	51.97	78.19	83.38
13	60.70	66.32	71.23	65.12	84.39	91.44	60.21	68.63	56.88	47.02	76.14	**89.12**
14	81.78	80.32	91.09	99.31	99.76	98.87	100.00	98.38	98.26	98.38	98.38	**100.00**
15	91.12	93.40	96.62	98.12	72.09	99.58	99.66	96.64	97.15	98.31	**100.00**	93.87
OA	74.99***	74.87***	83.12***	81.19***	86.75***	90.43	77.98***	82.31***	81.92***	78.22***	87.72***	**90.57**
AA	76.90***	77.26***	84.71***	83.46***	87.18***	91.77	79.97***	84.36***	83.55***	80.12***	88.83***	**91.51**
$\kappa \times 100$	72.89***	72.83***	81.69***	79.60***	85.61***	89.61	76.11***	80.84***	80.39***	76.39***	86.67***	**89.76**

TABLE VIII

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS ON THE KSC DATASET. "*," "**," AND "***" INDICATE THAT THE IMPROVEMENT OF THE CDCFORMER OVER THE BASELINE IS SIGNIFICANT AT THE 0.1, 0.05, AND 0.01 LEVELS, RESPECTIVELY

Class No.	KNN	RF	SVM	1-D CNN	2-D CNN	SSRN	HybridSN	RNN	SAE	ViT	SpectralFormer	CDCformer
1	**100.00**	**100.00**	**100.00**	90.64	99.50	96.18	92.16	88.55	93.15	91.27	98.76	98.89
2	57.58	75.97	80.95	86.10	**88.01**	79.65	62.38	74.68	84.11	63.42	82.25	75.32
3	70.78	40.26	84.77	83.33	**99.42**	39.92	81.32	82.67	74.69	36.38	76.13	89.30
4	37.66	97.16	46.86	35.15	**49.58**	29.46	43.81	13.31	41.76	4.27	40.59	46.44
5	7.84	56.03	11.11	45.42	46.73	49.93	77.19	13.73	48.56	4.58	**77.78**	57.52
6	69.27	46.47	86.24	65.64	87.39	34.95	69.04	46.74	47.20	0.78	85.78	**92.20**
7	16.00	86.19	78.00	57.5	**91.00**	39.4	52.5	37.40	74.60	0.	87.00	82.00
8	65.53	78.17	**92.91**	81.78	98.22	79.07	84.65	59.71	86.48	49.58	87.78	90.71
9	77.73	96.26	99.60	89.07	96.48	81.46	85.28	77.25	81.11	62.63	99.39	**100.00**
10	88.80	82.19	97.66	94.48	93.70	95.52	69.90	41.61	87.06	69.56	97.92	**98.44**
11	94.72	97.84	97.49	92.79	94.25	99.05	87.39	86.06	93.79	85.30	97.49	**100.00**
12	96.23	98.47	**99.79**	89.69	90.13	86.86	84.46	81.38	83.72	69.37	94.77	99.58
13	**100.00**	**100.00**	**100.00**	99.31	99.30	100.00	99.55	99.09	99.84	99.17	**100.00**	**100.00**
OA	80.95***	90.34***	90.97***	85.25***	92.10**	81.23***	82.78***	72.63***	83.67***	65.40***	91.44***	**92.63**
AA	67.86***	85.24***	82.72***	77.76***	**87.25**	70.11***	76.12***	61.71***	76.62***	48.95***	86.59***	86.95
$\kappa \times 100$	78.73***	89.23***	89.94***	83.59***	91.19**	79.06***	80.74***	69.49***	81.81***	60.85***	90.46***	**91.79**

good results when 20% of the training samples are involved, which largely demonstrates the robustness and effectiveness of the proposed CDCformer on a relatively small training set.

D. Classification Results and Analysis

1) Quantitative Analysis: Tables VII–IX report the quantitative classification results for the three overall indices, namely, OA, AA, and $\kappa \times 100$, for the Houston2013, KSC, and Botswana HS datasets, respectively, as well as the accuracy rates for each category. The best values are shown in bold. We can see that the proposed CDCformer implementation has the best OA, AA, and kappa values, with OA of 90.57%, 92.63%, and 97.57% on Houston2013, KSC, and Botswana, respectively. Among them, the training and test samples for Houston2013 were provided by the 2013 IEEE GRSS Data Fusion Competition, while the training sets for KSC and Botswana were randomly selected from 5% of the total samples.

In addition, to investigate the significant differences between the CDCformer and baseline models, we conducted the paired t-test for both the CDCformer and baseline models. The results of the significant tests are shown in Tables VII–IX, where "*," "**," and "***" indicate that the improvement of the CDCformer over the baseline is significant at the 0.1, 0.05, and 0.01 levels, respectively. From Tables VII–IX, we can observe that the CDCformer model significantly outperforms most of the models on the same testing set at the 0.01 level. The CDCformer model significantly outperforms the 2-D CNN model at the 0.05 level for OA and kappa. Similarly, the CDCformer model significantly outperforms the SpectralFormer model at 0.05 or 0.1 level. Unfortunately, the CDCformer model has no significant advantage over the SSRN model on the Houston2013 dataset. However, the CDCformer model significantly outperformed the SSRN model at the 0.01 level on the other two datasets. Thus, in general, the CDCformer model significantly outperformed these baseline models in the vast majority of cases. Namely, the CDCformer method is generally robust for these three datasets.

For the proposed CDCformer, the OA is 15.58% and 15.70% higher than KNN and RF, respectively, and 12.35%

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TABLE IX

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS ON THE BOTSWANA DATASET. "*," "**," AND "***" INDICATE THAT THE IMPROVEMENT OF THE CDCFORMER OVER THE BASELINE IS SIGNIFICANT AT THE 0.1, 0.05, AND 0.01 LEVELS, RESPECTIVELY

Class No.	KNN	RF	SVM	1-D CNN	2-D CNN	SSRN	HybridSN	RNN	SAE	ViT	SpectralFormer	CDCformer
1	96.30	92.63	96.71	**100.00**	98.81	100.00	92.43	99.67	**100.00**	**100.00**	**100.00**	99.59
2	82.42	98.90	**100.00**	77.25	75.93	93.30	98.46	77.14	76.59	86.81	96.70	97.80
3	95.13	95.13	**100.00**	90.80	98.94	99.51	98.72	93.41	96.64	93.81	**100.00**	99.56
4	**100.00**	96.96	**100.00**	83.45	99.79	97.94	99.59	84.48	88.56	84.54	**100.00**	97.42
5	55.79	71.86	72.31	82.52	83.43	89.67	90.74	81.45	85.00	89.67	85.95	91.32
6	91.32	80.91	97.52	76.28	91.20	82.15	80.33	66.65	74.50	59.50	91.32	92.15
7	98.71	99.57	**100.00**	97.17	99.36	99.83	100.00	97.30	97.77	94.42	**100.00**	**100.00**
8	86.89	95.74	**100.00**	92.35	99.40	97.70	95.08	93.01	94.86	92.90	**100.00**	**100.00**
9	97.17	93.14	**97.53**	82.97	95.27	92.58	98.30	79.08	83.04	61.84	96.11	95.76
10	**100.00**	99.78	**100.00**	63.14	99.82	93.09	98.83	76.37	80.81	64.13	**100.00**	99.10
11	98.18	97.16	97.45	90.84	94.62	96.95	99.94	90.91	93.45	92.73	**100.00**	99.64
12	85.89	90.18	90.80	89.75	89.02	97.79	99.88	92.82	95.71	90.80	98.16	**98.77**
13	99.17	99.09	**100.00**	81.04	99.67	96.89	88.00	75.81	84.11	73.86	99.59	97.92
14	80.00	82.12	75.29	99.88	93.06	94.12	99.89	97.76	98.12	**100.00**	90.59	**100.00**
OA	91.59***	92.48***	95.52***	85.91***	95.05***	95.04***	95.87***	85.63***	89.08***	83.21***	97.23*	**97.57**
AA	90.50***	92.37***	94.83***	86.25***	94.16***	95.11***	95.73***	86.13***	89.23***	84.64***	97.03**	**97.79**
$\kappa \times 100$	90.88***	91.85***	95.15***	84.73***	94.63***	94.63***	95.52***	84.43***	88.16***	81.82***	97.00*	**97.37**

and 2.85% better than ViT and SpectralFormer, respectively, on the Houston2013 dataset, for example. Moreover, the OA of CDCformer is improved by 2.85%, 1.19%, and 0.34% on the Houston2013, KSC, and Botswana HS datasets, respectively, compared with SpectralFormer. The experimental results show that the proposed CDCformer is advantageous in extracting the sequential information of HSI.

In particular, Table VII demonstrates that there are a couple of unbalanced classification results in detail. For the Houston2013 dataset, the "Commercial" and "Highway" categories perform poorly in the comparison methods. Since the Houston2013 dataset is mostly discrete and localized, unlike the other datasets, most of the regions belong to the same category. Therefore, our method has enough advantages in modeling the Houston2013 dataset to achieve very competitive results. It is worth noting that the SSRN model achieves impressive results on the Houston2013 dataset, but it has poor results on the other two datasets.

In addition, the "Slash pine" and "Hardwood swamp" categories in the KSC dataset and the "Hippo grass" and "Exposed soils" categories in the Botswana dataset have poor classification results for KNN, SVM, RNN, and SAE models. Since these two datasets are randomly sampled based on percentages, the sample sizes for these categories are quite small, resulting in relatively small numbers of training and imbalance samples for these categories. However, the accuracy achieved by CDCformer is relatively uniform in each category, which validates the effectiveness of the method for classifying small samples.

Using the spectral–spatial features generated by CCFT as input tokens for ViT, the proposed CDCFormer introduces some of the inductive bias inherent to CNNs into the transformer while maintaining the advantages of a transformer. In addition, the DCT enhances the information exchange between the layers, thus effectively improving the performance of the transformer model and extracting more discriminative information from the data. Therefore, due to the joint consideration of local spatial–contextual information and global spectral information in the feature extraction process, CDCformer achieves higher classification accuracy, which is far better than its competitors.

2) Visual Evaluation: Figs. 8–10 show the training, testing, ground-truth samples and the classification maps of each comparison model for the three datasets. By visual comparison, CDCformer's classification map is the cleanest and closest to the ground truth.

For these three datasets, the 1-D CNN, RNN, SAE, and ViT methods tend to produce salt and pepper noise on the classification maps, while, for KNN, RF, SVM, 2-D CNN, SSRN, and HybridSN classification models, the classification maps obtained are smooth at the edges, which leads to misclassification of pixels at the edges. The SpectralFormer model extracts highly continuous features from HSIs, leading to visualized classification map comparable to the classical networks mentioned above. In particular, our approach has fewer noisy pixels in comparison with those pixel-level approaches, such as 1-D CNN, RNN, SAE, and ViT, yet avoids oversmoothing of edges or few tiny semantic objects, such as KNN, RF, SVM, and 2-D CNN, resulting in more accurate classification performance with more realistic and fine details.

Therefore, by introducing some inductive bias inherent to CNNs into the transformer and enhancing feature propagation, our proposed CDCformer obtains very promising classification maps, especially for edge details.

3) Time Consumption: The execution times of the various methods are presented in Table X. The exercises were performed on an Intel Core i9-12900K processor (3.19 GH), 64-GB memory, and a GeForce RTX 3090 GPU. For traditional methods, including KNN and RF, the processing time is short, but the performance of such methods is poor. In addition, for SAE, SVM, and 1-D CNN, the running time is shorter compared to the other deep learning-based methods because these networks have fewer layers and contain fewer parameters. In addition, 2-D CNN, RNN, SSRN, and HybridSN models take longer time because more parameters are needed for training. As for the transformer-based ViT and SpectralFormer, the processing time is relatively longer than the other deep models because the proposed method considers

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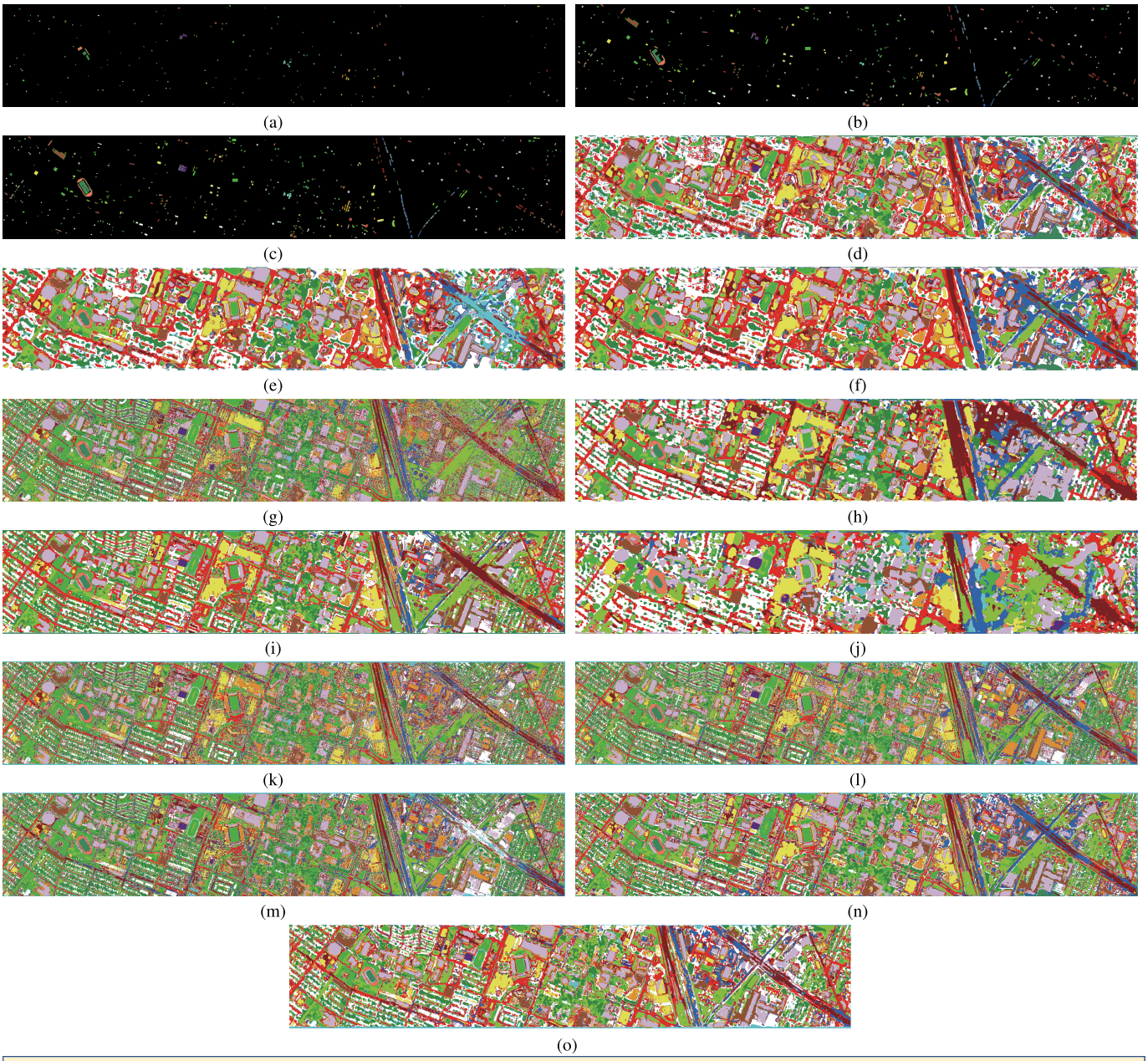


Fig. 8. Spatial distribution of the training and testing sets, and the classification maps acquired for different models on the Houston2013 dataset. (a) Training. (b) Testing. (c) Ground truth. (d) KNN. (e) RF. (f) SVM. (g) 1-DCNN. (h) 2-DCNN. (i) SSRN. (j) HybridSN. (k) RNN. (l) SAE. (m) ViT. (n) SpectralFormer. (o) CDCformer.

the transformer model. For the proposed CDCformer method, the processing time is longer than SpectralFormer because the proposed method performs a cascaded 3-D/2-D convolution network before feeding into the transformer encoder blocks.

4) Comparative Experiments With ViT and CNN Fusion Methods: Recently, many new methods for HSI classification based on fusion ViT and CNN have been proposed. Here, we add comparative experiments with current state-of-the-art fusion methods, which include CTMixer [52], SSFTT [53], GAHT [54], and HiT [55], as follows.

1) For CTMixer, a grouped parallel residual block is first applied to capture local spectral–spatial features in HSI patches. Second, a two-branch structure consisting of CNN and transformer is developed to capture local–global HS features. Finally, a novel local–global multiheaded SA mechanism (MHSA) is proposed to

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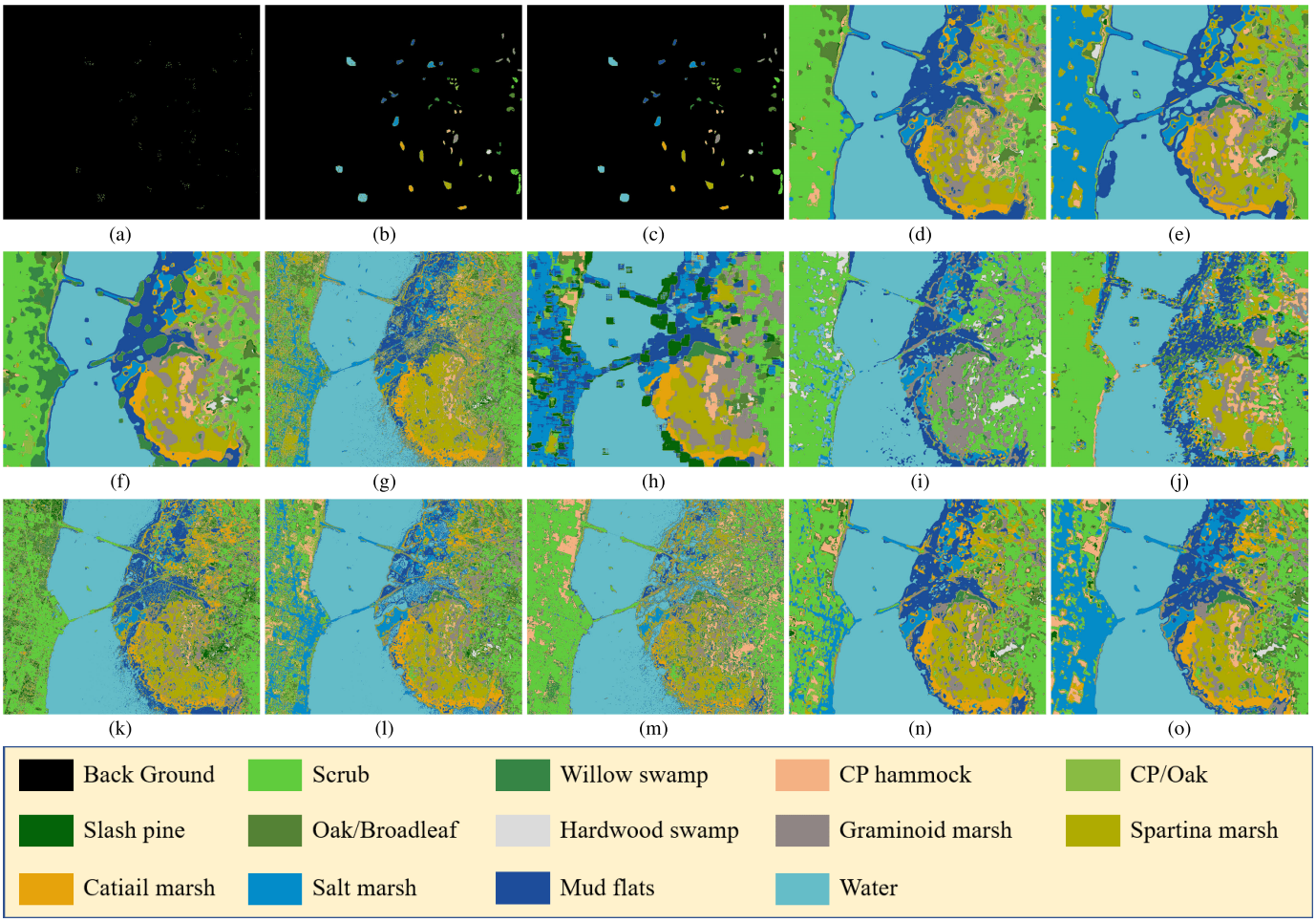


Fig. 9. Spatial distribution of the training and testing sets, and the classification maps acquired for different models on the KSC dataset. (a) Training. (b) Testing. (c) Ground truth. (d) KNN. (e) RF. (f) SVM. (g) 1-DCNN. (h) 2-DCNN. (i) SSRN. (j) HybridSN. (k) RNN. (l) SAE. (m) ViT. (n) SpectralFormer. (o) CDCformer.

TABLE X
TRAINING TIME AND TEST TIME IN SECONDS (S) ON THREE DATASETS

Methods	Houston2013		KSC		Botswana	
	Train	Test	Train	Test	Train	Test
KNN	0.001	0.55	0.0	0.06	0.0	0.34
RF	9.09	1.19	0.97	0.27	1.13	0.20
SVM	53.38	0.65	0.94	0.05	1.41	0.03
1-DCNN	41.48	0.01	5.46	0.01	5.91	0.005
2-DCNN	280.37	0.19	31.09	0.10	27.51	0.05
SSRN	749.08	0.99	117.32	0.48	188.32	0.48
HybridSN	369.37	1.04	101.55	0.43	79.13	0.26
RNN	182.64	0.05	24.75	0.02	16.42	0.01
SAE	19.87	0.002	2.88	0.001	2.78	0.001
ViT	439.68	0.65	93.90	0.32	66.02	0.16
SpectralFormer	644.51	1.03	123.39	0.50	94.01	0.27
CDCformer	974.33	1.32	180.24	0.65	137.78	0.31

further improve the classification accuracy by introducing convolution operations in MHSA [52].

2) The SSFTT method is proposed to capture spectral–spatial features and high-level semantic features. First, a spectral–spatial feature extraction module is built to capture the low-level features. Second, a Gaussian-weighted feature tokenizer is employed for feature transformation [53].
3) The GAHT method confines MHSA to the local spatial–spectral context by introducing a new GPE module. In addition, GAHT constructs the transformer in a hierarchical manner, which can significantly improve the classification accuracy [54].
4) The HiT classification network captures the subtle spectral discrepancies and conveys local spatial context information by embedding convolution operations into the transformer structure. HiT consists of two key modules, i.e., the spectral–adaptive 3-D convolution projection module and the convolution permutator (ConV-Permutator), to retrieve subtle spatial–spectral discrepancies [55].

The classification results of the state-of-the-art fusion methods for the three HSI datasets are shown in Table XI. From Table XI, we can observe that our CDCformer performs relatively well on the Houston2013 and KSC datasets, and slightly worse on the Botswana dataset. Although the CTMixer, SSFTT, and GAHT methods performed well on Houston2013 and Botswana datasets, they perform poorly on the KSC dataset compared to our CDCformer method. Therefore, the CDCformer method is generally robust for these three datasets.

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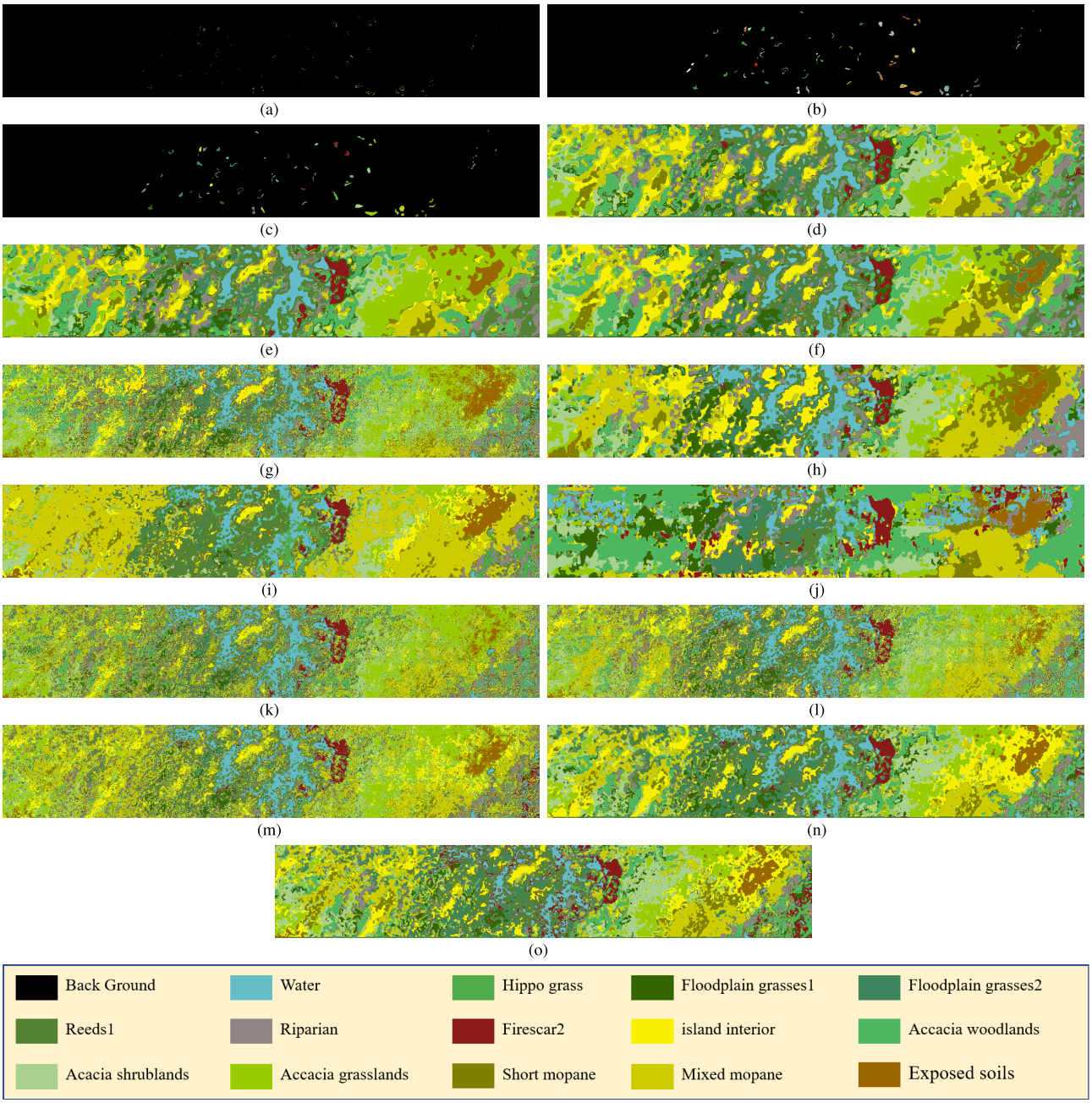


Fig. 10. Spatial distribution of the training and testing sets, and the classification maps acquired for different models on the Botswana dataset. (a) Training. (b) Testing. (c) Ground truth. (d) KNN. (e) RF. (f) SVM. (g) 1-DCNN. (h) 2-DCNN. (i) SSRN. (j) HybridSN. (k) RNN. (l) SAE. (m) ViT. (n) SpectralFormer. (o) CDCformer.

TABLE XI
COMPARATIVE EXPERIMENTS WITH STATE-OF-THE-ART FUSION METHODS ON THESE THREE DATASETS

Data sets	Metric	CNN and ViT Fusion Methods				
		CTMixer	SSFTT	GAHT	HiT	CDCformer
Houston –2013	OA (%)	90.32	90.45	90.20	81.43	**90.57**
	AA (%)	91.39	91.26	91.44	79.32	**91.51**
KSC	OA (%)	85.84	33.57	89.29	83.74	**92.63**
	AA (%)	78.78	24.06	81.31	71.15	**86.95**
Botswana	OA (%)	97.98	98.31	**98.93**	97.45	97.57
	AA (%)	98.13	98.36	**99.05**	97.16	97.79

IV. CONCLUSION

The classical ViT lacks the excellent ability of CNNs to establish local relationships in images. In addition, the gradient disappearance problem hinders the convergence stability of deep learning models as the number of layers goes deeper. To address these problems, this article proposes a CDCformer model for spectral–spatial HSI classification. The model organically combines CNN and transformer structures. The convolution layers are designed to fully capture both the spectral and spatial features, which could introduce some inductive bias inherent to the CNNs into the transformer. In addition, DCT is proposed to enhance feature propagation and provide memorable information during layer-by-layer propagation. Such operation effectively enhances the performance of the transformer model and extracts more discriminatory information from the data. Experiments show that this approach can effectively improve classification performance.

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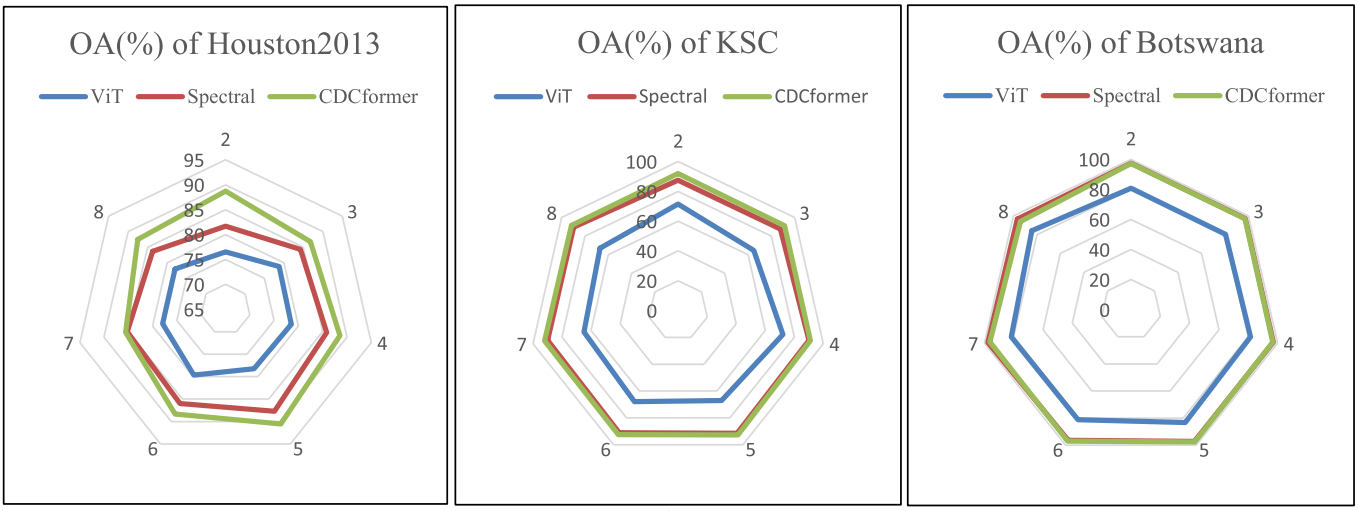


Fig. 11. OA with the varying depth number of the transformer encoder of ViT-based models on these three datasets. (a) Houston2013. (b) KSC. (c) Botswana.

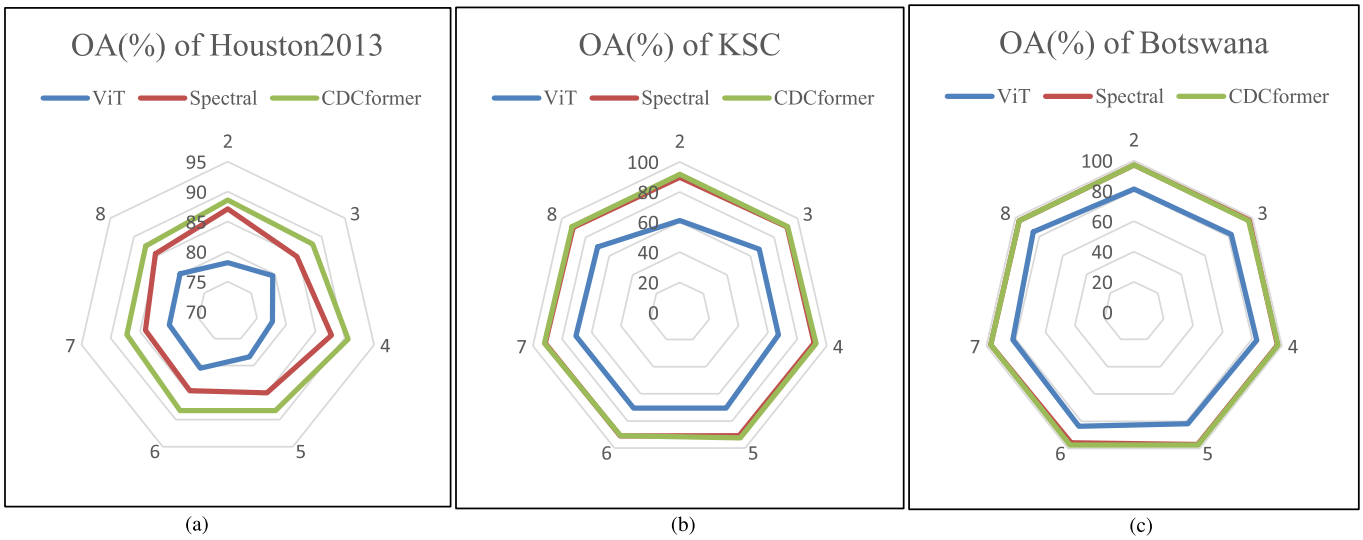


Fig. 12. OA with the varying number of MSA heads of the transformer encoder of ViT-based models on these three datasets. (a) Houston2013. (b) KSC. (c) Botswana.

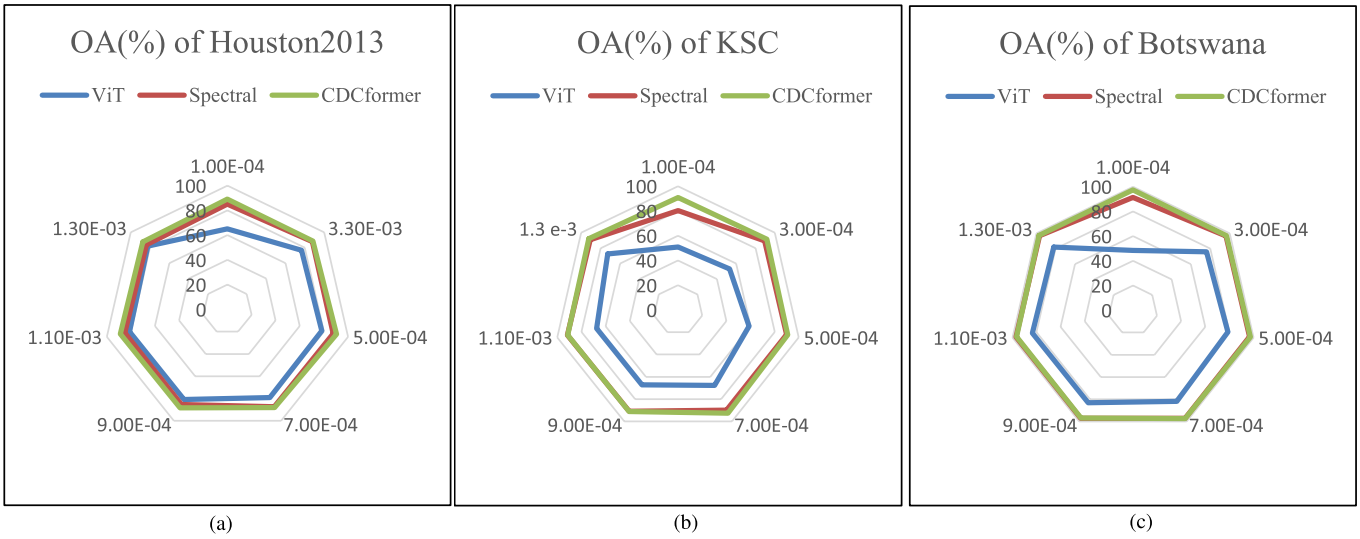


Fig. 13. OA with the varying learning rate of ViT-based models on these three datasets. (a) Houston2013. (b) KSC. (c) Botswana.

In the future, we will investigate strategies to improve the transformer structure by using advanced techniques, such as contrast learning, to make it more suitable for HSIs with minimal training datasets. In addition, we hope to build a

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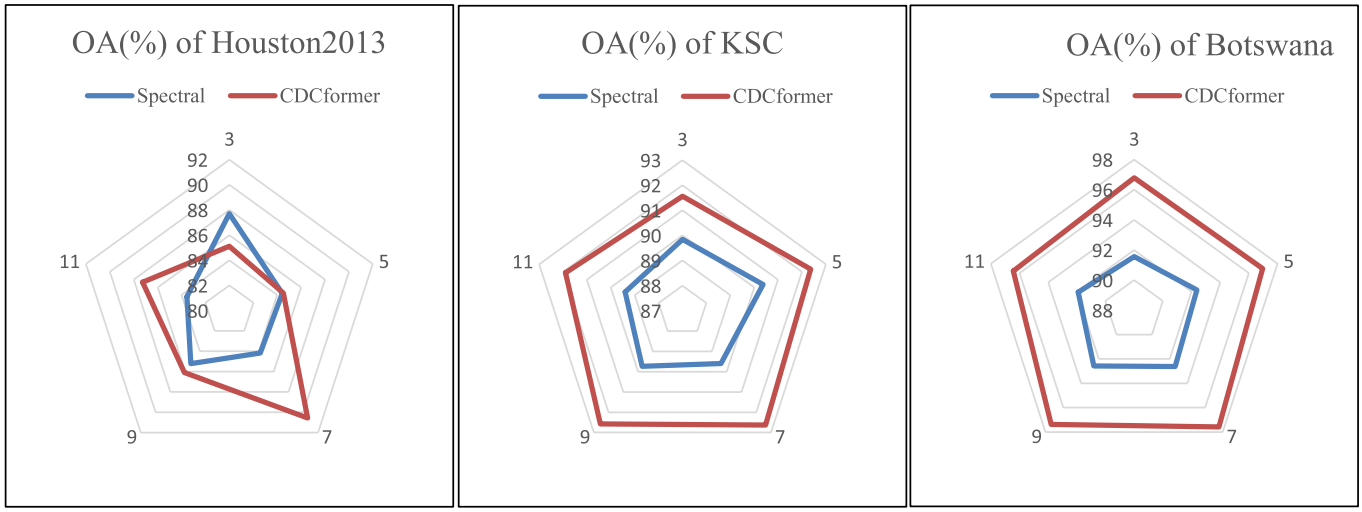


Fig. 14. OA with the varying number of adjacent bands of the SpectralFormer and CDCformer on these three datasets. (a) Houston2013. (b) KSC. (c) Botswana.

lightweight transformer-based network to achieve operational efficiency and reduce the complexity of the network while maintaining its performance.

APPENDIX

A. Some Results of Parameter Analysis

To demonstrate the effectiveness of the proposed method, we show the radar chart of the OA with the varying depth number of the transformer encoder of ViT-based models (see Fig. 11); OA with the varying number of MSA heads of the transformer encoder of ViT-based models in Fig. 12; OA with the varying number of learning rate of ViT-based models in Fig. 13; and OA with the varying number of adjacent bands of the SpectralFormer and CDCformer in Fig. 14 on these three datasets.

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